In class lab #5 for Numerical Analysis

Your name: _____

scan and submit before 23:00

Part I: Jacobi Method (40 points)

In order to rewrite $Ax = b$ into a form of $x = G(x)$, we should keep something to the left hand side and move everything else to the right hand side:

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{ii}\boxed{x_i} + \cdots + a_{in}x_n = b_i,\ 1 \le i \le n.$$

So we keep x_i in i-th equation to left hand side, and move everything else to the right hand side. Then we have

$$x_i = \left(b_i - \sum_{j=1, j\neq i}^{n} a_{ij}x_j \right) / a_{ii}.$$

It means we obtain $\vec{x} = G(\vec{x})$ with some linear function G. If $\vec{x}$ is the solution of $Ax - b = 0$, $\vec{x}$ is the fixed point. Here we assume that all diagonal entries are nonzero. If this is not the case, we can usually rearrange the equation so that it is.

The **Jacobi Method** is just the **fixed-point iteration**:

Step 1. Choose initial guess $x^{(0)} = (x_1^{(0)}, x_2^{(0)}, \cdots, x_n^{(0)})^T$.

Step 2. We do the following iterations

for $k = 0, 1, 2, \cdots, K-1$, where K is the number of iterations:

for $i = 1, 2, \cdots, n$

$$x_i^{(k+1)} = \left(b_i - \sum_{j=1, j\neq i}^{n} a_{ij}x_j^{(k)} \right) / a_{ii}.$$

end

end

We can choose any initial guess. We always choose $x^{(0)} = (0, 0, \cdots, 0)^T$.

Problem 1. (20 points) Apply two steps of Jacobi Method with initial guess x0 = 0.

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \tag{1}$$

Problem 2. Step 1 (5 points) Click on New to create a new document and type the code below. Once you are done, save it by the name `jacobi_1.m`.

```
% Program Jacobi Method
% Inputs: matrix A, right-hand side b,
%         initial guess x0,
%         the tolerance tol,
%         the maximum number of iterations maxIt
% output: solution x
function [x,k,err] = jacobi_1(A,b,x0,tol,maxIt)
n = size(A,1);
x = x0;
for k = 1:maxIt
for i = 1:n
tmp=0;
for j = 1:n
if( j==i )
tmp = tmp;
else
tmp = tmp + A(i,j)*x(j);
end
end
x1(i) = (b(i)-tmp)/A(i,i);
end
err=norm(x-x1,2);
if (err<tol)
x = x1;
break;
end
x = x1;
end

end
```

Step 2 (10 points) Use the `jacobi_1.m` to solve the equation (1) with $x_0 = 0$ within $tol = 10^{-8}$. Answer with results, the number of iterations and the error.

Step 3 (5 points) Use Matlab code $x = A \setminus b$ to solve the equation (1) and write the results you obtained.

Part II: Gauss–Seidel Method (60 points)

Closely related to the Jacobi Method is an iteration called the Gauss-Seidal Method. Note that Gauss-Seidal often **converges faster** than Jacobi if the method is convergent.

In the Jacobi method, the equations are solved in order of $i = 1, 2, \cdots, n$,

$$x_i^{(k+1)} = \left(b_i - \sum_{j=1, j\neq i}^{n} a_{ij} x_j^{(k)} \right) / a_{ii}.$$

For i-th equation, we have already got $x_j^{(k+1)}, j = 1, \cdots, i-1$. We clearly know that $x_j^{(k+1)}$ is more accurate than $x_j^{(k)}, j = 1, \cdots, i-1$. So taking into account of this, we change the Jacobi Method

$$x_i^{(k+1)} = \left(b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k)} - \sum_{j=i+1}^{n} a_{ij} x_j^{(k)} \right) / a_{ii}$$

to Gauss-Seidal Method

$$x_i^{(k+1)} = \left(b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i+1}^{n} a_{ij} x_j^{(k)} \right) / a_{ii}.$$

Problem 3. (20 points) **Complete the code in the box** to make it a **Gauss–Seidel method**. Click on New to create a new document and type the code. Once you are done, save it by the name `gauss_seidel_1.m`.:

```
% Program Gauss-Seidel Method
% Inputs: matrix A, right-hand side b,
%         initial guess x0,
%         the tolerance tol,
%         the maximum number of iterations maxIt
% output: solution x
function [x,k,err] = gauss_seidel_1(A,b,x0,tol,maxIt)
n = size(A,1);
x = x0;
for k = 1:maxIt
for i = 1:n
tmp=0;
for j = 1:n
if( j==i )
tmp = tmp;
else
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-------------------------------------------------------------------------
```

```
end
end
x1(i) = (b(i)-tmp)/A(i,i);
end
err=norm(x-x1,2);
if (err<tol)
x = x1;
break;
end
x = x1;
end
end
```

Problem 4. (40 points) Consider the following differential equation:

$$\begin{cases} \varepsilon \dfrac{\mathrm{d}^2 y}{\mathrm{d}x^2} + \dfrac{\mathrm{d}y}{\mathrm{d}x} = a, \quad 0 < a < 1 \\ y(0) = 0, y(1) = 1 \end{cases}$$

whose exact solution is

$$y_* = \frac{1-a}{1-\mathrm{e}^{-\frac{1}{\varepsilon}}} \left(1 - \mathrm{e}^{-\frac{x}{\varepsilon}}\right) + ax.$$

Divide the interval $[0, 1]$ into n equal parts,

$$x_i = ih, \quad h = 1/n, \quad i = 1, \cdots, n-1,$$

and we obtain the difference equation

$$\varepsilon \frac{y_{i-1} - 2y_i + y_{i+1}}{h^2} + \frac{y_{i+1} - y_i}{h} = a,$$

which can be simplified as

$$(\varepsilon + h)y_{i+1} - (2\varepsilon + h)y_i + \varepsilon y_{i-1} = ah^2.$$

After discretization, the coefficient matrix of the linear system is

$$A = \begin{bmatrix} -(2\varepsilon+h) & \varepsilon+h & & & \\ \varepsilon & -(2\varepsilon+h) & \varepsilon+h & & \\ & \varepsilon & -(2\varepsilon+h) & \ddots & \\ & & \ddots & \ddots & \varepsilon+h \\ & & & \varepsilon & -(2\varepsilon+h) \end{bmatrix}.$$

Setting $\varepsilon = 1$, $a = 1/2$, $n = 100$, initial guess $\mathbf{y}_0 = 0$, calculate the solutions via the Jacobi method and the G-S method, respectively, within $tol = 10^{-8}$. Report the number of iterations, the backward error, the forward error, and $cond(A)$. Repeat the work with $\varepsilon = 0.1$, $\varepsilon = 0.01$, $\varepsilon = 0.001$, and $\varepsilon = 0.0001$.